

Birinder Bachhal

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Education

Boston University

Kilachand Honors College/College of Engineering

Intended Degree: Mechanical Engineering

Awards/Organizations: F1Tenth ACRO, Sikh Association at BU, Theta Tau Professional Engineering Fraternity, QuestBridge Scholar, Society of Asian Scientists and Engineers

Boston, MA

September 2024 – May 2028

Technical Skills

Computer: SolidWorks, AutoCAD, FEA, Python, MATLAB, Arduino, Reinforcement Learning, Microsoft Suite

Mechanical: Additive Manufacturing, DFM, GD&T, Thermal Analysis, Waterproofing, CNC Machining, Drill Press, Laser Cutter, CAM

Languages: English, Punjabi, Hindi, Urdu, Spanish

Work Experience

Deteqt

Sydney, Australia

Mechanical Engineering Intern

January 2026 – Present

- Designed ruggedized IP65-rated enclosure in SolidWorks for 24-hour field deployment in $>40^{\circ}\text{C}$ environments, completing CAD models, GD&T technical drawings, and 4 rapid prototyping iterations via FDM and SLA 3D printing
- Engineered aluminum baseplate heatsink to dissipate 2-5W thermal load while maintaining quantum sensor performance, applying thermal modeling and DFM principles for CNC manufacturing
- Prepared production-ready designs, standardizing fasteners, integrating heat-set brass inserts, and delivering complete technical documentation needed for outsourced manufacturing
- Built and tested waterproof fixtures including flush-mount USB-C housing and O-ring seals, ensuring hardware compatibility and environmental protection

Boston University Chemistry Department: General Chemistry

Boston, MA

Lab Assistant

September 2024 – June 2025

- Effectively manage the instructional laboratories in the Chemistry Department by setting up the laboratory assignments before each section
- Deliver supplies, manage chemical inventory, organize and report hazardous waste, and prepare reports regarding laboratory standards

Projects

BU F1Tenth Autonomous Race Car

Boston, MA

Executive Board Member: Hardware

December 2024 – Present

- Leading mechanical design on a Guinness World Record attempt autonomous car, converting a 2008 Corvette into an autonomous platform
- Designed chassis components for 1:10 scale prototype, improving structural rigidity by 30% and reducing weight by 15% through CAD
- Integrated LIDAR, servo control, and sensor systems, achieving 98% hardware-software communication reliability in bench tests
- Modeled and validated vehicle dynamics in CARLA and ROAR simulators, enhancing path-planning accuracy by 25% in simulation trials

OrthoCAD Pro

Boston, MA

Personal-Interest Project

July 2025 – September 2025

- Developed a web-based orthotic design platform integrating computer vision and parametric CAD to automate custom orthotic fabrication
- Implemented a JavaScript-based photogrammetry pipeline achieving a ± 1 mm precision, exceeding clinical tolerance by 50% for smartphone-based 3D foot scans
- Automated CAD generation in Fusion 360 and SolidWorks, producing manufacturing-ready orthotic models with customizable parameters

The Deskinator – Autonomous Workspace Robot

Boston, MA

Design Lead

September 2025 – Present

- Designed an autonomous desktop-cleaning robot using pose-graph estimation and sensor fusion for live edge detection and navigation
- Integrated NEMA-17 stepper motors, proximity sensors, and custom-machined housing components to achieve $> 95\%$ edge-tracking accuracy and smooth, controlled motion
- Developed embedded control logic on Raspberry Pi with PID feedback, improving motion stability, and manufacturability by 40% across 3 prototype iterations

Academic Experience

First-Year Innovation Fellowship

Boston, MA

2024-2025 Fellow

September 2024 – June 2025

- Designed 2 prototypes for an insulin delivery system integrating mechanical design with embedded software modeling pancreas behavior
- Applied machine learning concepts and control design to biomedical hardware for real-time glucose management

Leadership

Executive Board Member: BU F1Tenth

December 2024 – Present

Executive Board Member: Sikh Association at Boston University

June 2025 – Present

Academic Chair: Theta Tau Professional Engineering Fraternity

June 2025 – January 2026